Recall Radar – Product Requirements Document

Automated detection of dangerous products on online marketplaces under the EU General Product Safety Regulation

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Document control

Item	Value
Product	Recall Radar (compliance workspace) and RadarMart (demonstration marketplace)
Document type	Product Requirements Document (PRD)
Owner	Abhinandan Samal (Product Owner)
Status	Draft for approval, Sprint 0 (Product Discovery)
Version	1.0, 8 November 2026
Scope	What the product solves, for whom, with which features and which measurable value. Technical architecture is documented separately (HLD/LLD, ADRs).

Change history

Version	Date	Change
0.1	08.11.2026	First concept: check each Safety Gate alert against a catalogue

Version	Date	Change
0.2	08.11.2026	Discovery finding: manual study showed the demo catalogue rarely contains alerted products. Product extended to check every **listing** against all known dangerous products (two triggers)
1.0	08.11.2026	Complete PRD including the RadarMart demonstration marketplace

1. Executive summary

Online marketplaces in Germany handle more than half of all online retail. Since December 2024 the EU General Product Safety Regulation (GPSR) makes them responsible for acting quickly against dangerous products, and since February 2026 Germany can fine breaches. At the same time, the EU's rapid alert system for dangerous products (Safety Gate) recorded a record number of alerts in 2025, and the Commission's own web crawler found tens of thousands of already-alerted products still offered online.

Marketplaces cannot solve this with keyword search. Alerts often have no barcode or brand, are written in another language than the listing, and the same product is re-uploaded under new names by new sellers.

Recall Radar checks every new or changed listing, and the whole catalogue whenever a new alert is published, against all known dangerous products. It compares **photos, meaning across languages and identifiers**, explains each finding in plain language with evidence and legal citations, and lets a human approve every action (listing removal, seller notice, buyer recall notice). Every step is recorded for the authorities.

RadarMart is a fully working demonstration marketplace (storefront, seller portal, cart, checkout, orders) connected to Recall Radar the same way a real marketplace would be, so the complete loop can be shown end to end: a seller uploads a dangerous product, Recall Radar detects it within seconds, a reviewer approves, the listing disappears and buyers who already ordered it are notified.

2. Problem statement

2.1 The problem in one sentence

Online marketplaces must find and remove listings of products that EU authorities have identified as dangerous, quickly and verifiably, but the manual process is slow, incomplete and cannot keep up with the volume, the languages and the re-uploads.

2.2 Market context (facts)

Fact	Figure	Source
German e-commerce goods revenue 2025	€83.1 bn (+3.2 %)	bevh [1]
Share sold via online marketplaces	56 % (€46.2 bn, +4.9 %)	bevh [1]
Safety Gate alerts in 2025	4,671, the highest since 2003; +13 % vs. 2024, more than double 2022	European Commission via [2]

Fact	Figure	Source
Most notified categories 2025	Cosmetics 36 %, toys 16 %, electrical appliances 11 %	[3]
Alerts notified by Germany in 2025	465 (second-highest)	[4]
Already-alerted products found still offered online in 2025	more than 20,800 (Commission eSurveillance web crawler)	[2]
Products from 7 foreign marketplaces failing EU rules (France, 2025, 600+ tested)	75 %; 46 % also dangerous	[5]

2.3 Legal pressure on marketplaces

- **GPSR Article 22:** marketplaces must register on the Safety Gate portal with a single point of contact, run internal product-safety processes, act on authority orders **within two working days** and on product-safety notices **within three working days**, and inform consumers about recalls [6][7].
- **DSA Article 31:** after admitting a trader, the marketplace must make reasonable efforts to **randomly check official databases** (such as Safety Gate) for products identified as illegal [8].
- **Germany:** since 19 February 2026, the amended Product Safety Act (ProdSG) provides fines of up to €10,000, and up to €100,000 for serious breaches, per breach [9].
- **Compliance gap:** in the 2025 EU sweep only 53 % of checked marketplaces were registered on Safety Gate, and 66 to 70 % answered authority requests within two working days [10].

2.4 Who is affected

Persona	Role	Pain today
Lena, Product-Safety Analyst (primary user)	Screens alerts and listings	~90 new alerts every week, a large catalogue, several languages; keyword search misses most cases
Markus, Compliance Reviewer (approver)	Decides takedowns and notices	Must decide within 2 to 3 working days and later prove why
Compliance Lead / Management	Owns GPSR compliance	No overview of exposure, response times or open risks; fine risk
Sellers	List products	Blocked without explanation, or never warned
Buyers	Buy products	Receive dangerous products that were already known to be dangerous

3. How the manual process works today, and why it fails

3.1 How products get onto a marketplace

Most products on marketplaces are offered by independent sellers. The marketplace verifies the **seller's identity** (DSA Article 30) and runs automatic checks on the **listing data** (required fields, banned words). It does not physically inspect products, and the law does not oblige it to monitor everything (DSA Article 8); instead it must check official databases at random (DSA Article 31) [8]. Listings typically go live within minutes.

3.2 Challenges of manual work

Challenge	Evidence
Missing identifiers	In 716 recent alerts, only 52 % had a barcode and 66 % a brand; photos were present in 100 % (own analysis, Nov 2026)
Alerts describe products, not listings	Only 677 of 8,739 alerts in the last two years name the marketplace where the product was sold (own analysis)
Language and wording differ	Alerts come from 30 countries; the same item is a “laser projector” in the alert and a “Party Licht Disco Lampe” in the listing
Re-uploads and look-alikes	Removed products reappear under new names or sellers [11][12]; 443 of 8,739 alerts are repeats of the same product (own analysis)
Different category systems	Marketplace product types (e.g. “BEAUTY”) do not map to Safety Gate categories (e.g. “Cosmetics”) (own manual study)
Volume and deadlines	~90 alerts per week, thousands of new listings per day, 2 to 3 working days to act
Proof	Decisions are scattered across e-mails and spreadsheets; hard to show authorities a complete record

3.3 Discovery evidence (own manual study, 8 Nov 2026)

The Product Owner tried to match real alerts against a marketplace catalogue by hand using name, category and model search. Most searches returned unrelated products or “no match”; category names did not match the catalogue’s types. This confirmed that keyword search is not a workable method, and led to the two-trigger design below. A second timed study (10 listings checked by hand against Safety Gate) establishes the baseline for the value measurement (Section 8).

4. Product vision, goals and non-goals

Vision. No product that EU authorities have identified as dangerous stays on sale on a marketplace longer than it takes a reviewer to click “approve”.

Goals

1. Detect known dangerous products among listings automatically, across photos, languages and identifiers.
2. Cut the time from alert publication (or listing upload) to a reviewed decision from days to minutes.
3. Keep humans in control of every consequential action, with clear, plain-language reasons.
4. Produce a complete, exportable record that demonstrates GPSR and DSA compliance.
5. Be connectable to any marketplace through a standard connector.

Non-goals

- No automatic takedown without human approval.
- No legal advice; legal texts are cited for information.
- No laboratory testing or product certification.

- No real marketplace data or real payments in the demonstration (RadarMart uses public data and simulated payment).

5. The solution

5.1 How it works (user view)

Step	What happens
1. Know	Recall Radar keeps an always-updated knowledge base of all published Safety Gate alerts (since 2005, English and German) and the GPSR text
2. Listen	It receives an event whenever a seller creates or changes a listing, and checks for new alerts every hour
3. Screen	**Trigger A, new or changed listing:** the listing is checked against all known dangerous products. **Trigger B, new alert:** the whole catalogue is checked for that product. A nightly run re-checks everything that changed
4. Explain	Each finding shows the photo comparison, matching details, risk, confidence and the legal duty in plain language, with citations
5. Decide	A reviewer approves or rejects proposed actions (remove listing, inform seller, notify buyers)
6. Act and record	Approved actions are executed in the marketplace; the team is notified; every step is logged

5.2 Result categories

Result	Meaning	What happens
Match	Very likely the same product as a known dangerous product	Case opened, team notified, actions proposed for approval
Similar	Same kind of product with notable resemblance	Case opened for human review, no action proposed until reviewed
Clear	No known dangerous product found	Recorded; no alert to the team

5.3 Plain-language explanations

Every result answers three questions in simple words for non-experts: **What did we find? Why do we think so? What must happen, by when?** Example:

> "This listing very likely shows the pacifier clip from EU alert SR/02809/26 (risk: strangulation). The photos match closely and the model number PS-001 is identical. Under the EU product safety rules the marketplace must remove such offers when ordered by an authority within two working days and inform buyers. Suggested actions: remove the listing, notify the seller, notify 3 buyers."

6. Product components and features

Priorities follow MoSCoW: **M** = must have for v1.0, **S** = should have (v1.0 if time allows, else v1.1), **C** = could have (later).

6.1 Recall Radar: compliance workspace

A. Knowledge base and monitoring

ID	Feature	Prio
A1	Automatic import of all Safety Gate weekly reports and alerts (EN + DE), including photos, risks and measures	M
A2	Hourly check for new alerts; history of updates and withdrawals per alert	M
A3	GPSR legal text (EN + DE), searchable by article	M
A4	Data quality checks on every import, with quarantine of invalid records	M
A5	Monitoring dashboard: last import, new alerts, data quality results	S

B. Screening

ID	Feature	Prio
B1	Listing screening on create/update event (Trigger A), result within seconds	M
B2	Catalogue back-scan when a new alert arrives (Trigger B)	M
B3	Nightly reconciliation of changed listings and changed alerts	M
B4	Matching by photo similarity, text meaning across languages, barcode/model	M
B5	Confidence score and result category (match / similar / clear)	M
B6	Random spot-check campaigns over the existing catalogue (DSA Art. 31)	S
B7	Configurable thresholds per category	S

C. Case management and human review

ID	Feature	Prio
C1	Case per finding with evidence: photos side by side, matched fields, alert details	M
C2	Plain-language explanation with legal citations (EN/DE)	M
C3	Review queue with approve / reject / request-more-information and mandatory reason	M
C4	Deadline timers (2 and 3 working days) and overdue warnings	M
C5	Four-eyes principle: analyst prepares, reviewer approves	S
C6	Comments and history per case	S

D. Actions and notifications

ID	Feature	Prio
D1	Remove listing in the marketplace after approval	M
D2	Seller notice (draft generated, approved, sent)	M
D3	Buyer recall notice to customers who ordered the product (draft, approval, sending)	M
D4	Team notifications via Discord and Gmail (new case, overdue case, daily summary)	M
D5	Undo / reinstate listing with reason	S

E. Compliance assistant (chat)

ID	Feature	Prio
E1	Chat in English and German about cases, alerts and GPSR duties, answers with citations	M
E2	“Explain like I am not a lawyer” answers by default	M
E3	Refuses to give legal advice and to act outside the user’s role	M

F. Audit, reporting and dashboards

ID	Feature	Prio
F1	Complete audit log: who did what, when, based on which evidence	M
F2	Export of case records (PDF/CSV) for authorities	S
F3	KPI dashboard: time to detection, time to decision, deadline compliance, false-alarm rate, cost per check	M

G. Administration and access

ID	Feature	Prio
G1	Separate logins and roles: Analyst, Reviewer, Admin	M
G2	User management and role assignment	M
G3	Connector management: marketplace connection, keys, health status	S
G4	English and German user interface	M

6.2 RadarMart: demonstration marketplace

RadarMart is a fictional marketplace built to production standards. It is the place where sellers list products and buyers shop, so that Recall Radar can be shown working on a live system. Its catalogue is built from public data (Amazon Berkeley Objects listings, CC BY 4.0) plus re-uploaded products from Safety Gate alerts.

H. Storefront (buyers)

ID	Feature	Prio
H1	Home page, category navigation, product search with filters (EN/DE)	M
H2	Product page with photos, price, description and GPSR information (manufacturer, EU responsible person, warnings)	M
H3	Shopping cart: add, change quantity, remove	M
H4	Checkout with address and **simulated payment** (no real money)	M
H5	Buyer account: registration, login, order history	M
H6	Product safety notice on removed products and recall messages in the buyer's account and by e-mail	M
H7	"Report a safety concern" button on every product page (feeds Recall Radar as a notice)	S
H8	Wish list, reviews	C

I. Seller portal

ID	Feature	Prio
I1	Seller registration with trader details (name, address, contact; DSA Art. 30 style verification, simulated)	M
I2	Create and edit listings: title, description, category, price, stock, photos, EAN, GPSR fields	M
I3	Listing status: live, under review, removed (with reason)	M
I4	Messages from the marketplace (safety notices)	M
I5	Bulk upload via CSV	C

J. Marketplace back office

ID	Feature	Prio
J1	Overview of listings and orders	M
J2	Every listing create/update sends an event to Recall Radar	M
J3	Executes approved actions from Recall Radar (removal, notices)	M

K. Security and privacy (both products)

ID	Requirement	Prio
K1	Separate roles: buyer, seller, marketplace admin (RadarMart); analyst, reviewer, admin (Recall Radar)	M
K2	Secure login, password hashing, session protection, CSRF protection, rate limiting	M

ID	Requirement	Prio
K3	Image upload validation (type, size), safe image fetching from allowed domains only	M
K4	Signed events between RadarMart and Recall Radar; duplicate events ignored	M
K5	Protection against prompt injection hidden in listing text	M
K6	GDPR: only demo personal data, data minimisation in logs, privacy notice, deletion on request	M
K7	AI transparency: users are told when content is AI-generated (EU AI Act Art. 50)	M

7. Key user journeys

1. **Re-upload caught at upload.** A seller lists a pacifier clip identical to an alerted product. Within seconds Recall Radar opens a case; the reviewer approves; the listing shows “removed – product safety”; the seller receives a notice.
2. **New alert, catalogue scan.** A new alert for a laser projector is published. Within the hour Recall Radar finds two RadarMart listings, notifies the team on Discord and by Gmail, and proposes removal.
3. **Buyers protected.** One of those listings had three orders. After approval, the three buyers get a recall notice in their account and by e-mail.
4. **Safe product passes.** A seller uploads a safe baby bib; the result is “clear” and no one is disturbed.
5. **Question to the assistant.** The reviewer asks in German which deadline applies; the assistant answers in plain language and cites the GPSR article.
6. **Audit.** The compliance lead exports all decisions of the last month for the authority.

8. Value and success measurement

8.1 Value drivers

Value	How Recall Radar creates it
Speed	Minutes instead of days from alert or upload to a reviewed decision
Coverage	Every alert is checked against every listing; every new listing is checked against every alert
Prevention	Re-uploads are caught before the first sale
Efficiency	Analysts review a short list of evidence-backed candidates instead of searching
Compliance and risk	Deadlines tracked, complete audit trail; lower risk of fines (up to €100,000 per breach in Germany) and of consumer harm
Trust	Buyers are informed about recalls; sellers receive clear reasons

8.2 KPIs

Targets marked *provisional* are confirmed after the baseline study.

KPI	Definition	Target (v1.0)
Catch rate	Share of known dangerous products in the test set that are detected (match or similar)	provisional ≥ 90 %

KPI	Definition	Target (v1.0)
False-alarm rate	Share of safe listings wrongly flagged	provisional ≤ 5 %
Time to detection	Event received → result available (listing check)	provisional p95 ≤ 10 s
Back-scan time	New alert → catalogue scan finished	provisional ≤ 5 min
Time to decision	Case opened → reviewer decision	measured; compared with baseline
Analyst minutes per check	Manual (baseline) vs. assisted	measured
Deadline compliance	Cases decided within 2 / 3 working days	100 % in demo
Explanation quality	Citation correctness and readability (Flesch / Flesch-Amstad)	provisional ≥ 95 % correct citations
Cost per 1,000 checks	Cloud + AI cost	measured; total running cost ≤ €30 per month

8.3 Baseline and proof

The baseline is measured by hand before development (Study 2: 10 listings, time per check, accuracy, confidence). The same cases are processed with Recall Radar after development. The difference is reported in the release notes, together with an extrapolation: minutes saved per check × number of checks per year. Example for orientation only: if a manual check took 30 minutes and an assisted one 8 minutes, the 4,671 alerts of 2025 alone would save about 1,700 analyst hours per year.

9. Product-level quality requirements

Area	Requirement
Availability	The live demo is reachable 24/7; it may take a few seconds to wake up after idle time (cost control)
Performance	See KPIs; the storefront page loads quickly on a normal connection
Languages	English and German throughout (UI, explanations, notices)
Accessibility	Keyboard navigation, sufficient contrast, alt texts
Reliability	Failed steps are retried automatically; failures are logged, explained and reported
Privacy	GDPR principles; demo data only
AI governance	Human approval for every consequential action; AI-generated text labelled; results explained
Cost	Total running cost ≤ €30 per month

10. Data sources

Source	Use	Licence
EU Safety Gate weekly reports (2005 to today, EN/DE)	Knowledge base of dangerous products; test cases	Commission reuse policy, attribution

Source	Use	Licence
Amazon Berkeley Objects (147,702 listings, 398,212 images)	RadarMart catalogue of normal, safe products	CC BY 4.0 (attribution)
GPSR, Regulation (EU) 2023/988 (EN/DE, EUR-Lex)	Legal duties, citations	EUR-Lex reuse, attribution

11. Assumptions, constraints and risks

Item	Type	Handling
Toys are the second-largest alert category but absent from the public catalogue	Risk	Toy detection is tested with Safety Gate data only; stated openly
No real marketplace traffic	Constraint	RadarMart and a traffic simulator generate realistic events
German full-text and photo matching quality unknown before testing	Risk	Every matching stage is evaluated on a labelled test set before release
Photos in alerts may belong to third parties	Risk	Linked and fingerprinted, not redistributed, until licence terms are confirmed
Budget ≤ €30 per month	Constraint	Serverless hosting, scale-to-zero, small open/efficient AI models
Laws change (e.g. EU AI Act timelines)	Risk	Legal texts are versioned; sources dated

12. Release plan

Release	Content
v1.0 (end of project)	All **M** features of Recall Radar and RadarMart, end-to-end demo, measured KPIs
v1.1	**S** features (spot-check campaigns, four-eyes principle, export, connector management, safety-concern button)
Later	**C** features, connectors for real marketplace APIs, more languages

The detailed sprint plan, backlog and progress are tracked on the project board and in the repository.

13. Glossary

Term	Meaning
Safety Gate	EU rapid alert system for dangerous non-food products
GPSR	General Product Safety Regulation (EU) 2023/988
DSA	Digital Services Act (EU) 2022/2065
Listing	A product offer by a seller on a marketplace
Alert	A published notification about a dangerous product

Term	Meaning
Case	A finding that needs human review
HITL	Human in the loop: a person approves before an action is executed

14. Sources

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